

Lab 1

Class and Object

Object Oriented Programming: is an approach or a programming pattern where the programs are structured around objects rather than functions and logic. It makes the data partitioned into two memory areas, i.e., data and functions, and helps make the code flexible and modular.

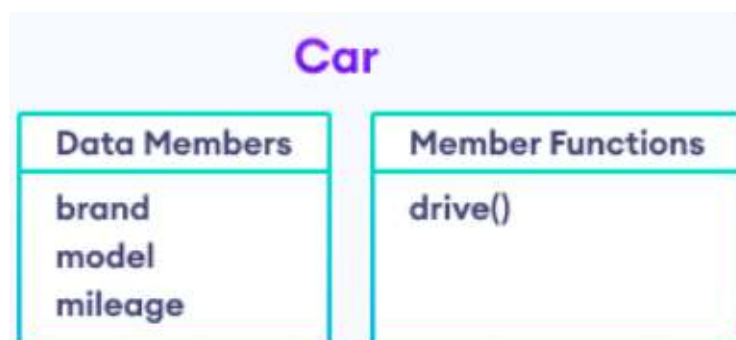
Class:

- A Class is a user-defined data type that has data members and member functions.
- Data members are the data variables and member functions are the functions used to manipulate these variables together these data members and member functions define the properties and behavior of the objects in a Class.

Object:

- An Object is an identifiable entity with some characteristics and behavior.
- An Object is an instance of a Class.
- When a class is defined, no memory is allocated but when it is instantiated (i.e. an object is created) memory is allocated.

Class	objects
Fruit	Apple Banana Mango



Example 1:

```
#include <iostream>
#include <string>
using namespace std;

class MyClass {          // The class
public:                  // Access specifier
    int myNum;           // Attribute (int variable)
    string myString;     // Attribute (string variable)
};

int main() {
    MyClass myObj;      // Create an object of MyClass

    // Access attributes and set values
    myObj.myNum = 15;
    myObj.myString = "Some text";

    // Print values
    cout << myObj.myNum << "\n";
    cout << myObj.myString;
    return 0;
}
```

Example 2:

```
#include <iostream>

using namespace std;

class Box {
public:
    double length;       // Length of a box
    double breadth;      // Breadth of a box
    double height;       // Height of a box
};

int main() {
    Box Box1;            // Declare Box1 of type Box
    Box Box2;            // Declare Box2 of type Box
    double volume = 0.0; // Store the volume of a
    box here

    // box 1 specification
    Box1.height = 5.0;
    Box1.length = 6.0;
    Box1.breadth = 7.0;
```

```
// box 2 specification
Box2.height = 10.0;
Box2.length = 12.0;
Box2.breadth = 13.0;

// volume of box 1
volume = Box1.height * Box1.length * Box1.breadth;
cout << "Volume of Box1 : " << volume <<endl;

// volume of box 2
volume = Box2.height * Box2.length * Box2.breadth;
cout << "Volume of Box2 : " << volume <<endl;
return 0;
}
```